

Detailed Notes on Matrix Multiplication

1. Introduction to Matrices

A **matrix** is a rectangular arrangement of numbers, symbols, or expressions in rows and columns.

Example:

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

This matrix has:

- **2 rows**
- **2 columns**
- Order (or dimension): 2×2

A general matrix with m rows and n columns is called an $m \times n$ matrix.

2. What is Matrix Multiplication?

Matrix multiplication is an operation where two matrices are combined to produce a third matrix.

If:

$$A \times B = C$$

then each element of matrix C is found by multiplying a **row of** A with a **column of** B and adding the results.

The rule is:

$$(AB)_{ij} = \sum_k a_{ik} b_{kj}$$

This means the element in row i and column j of the product matrix is obtained by taking the dot product of row i of A and column j of B .

3. Condition for Matrix Multiplication

Two matrices can be multiplied only when:

Number of columns in first matrix = Number of rows in second matrix

If:

$$A_{m \times n} \times B_{n \times p}$$

then the result will have dimensions:

$$C_{m \times p}$$

Example

Matrix A :

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Order:

$$2 \times 3$$

Matrix B :

$$B = \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$$

Order:

$$3 \times 2$$

Since:

$$(2 \times 3)(3 \times 2)$$

has matching inner dimensions, multiplication is possible.

The resulting matrix will have order:

$$2 \times 2$$

4. Steps for Matrix Multiplication

Consider:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

and:

$$B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

The product AB is:

$$AB = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$$

Explanation

First element:

$$(1, 1) = ae + bg$$

First row of A multiplied by first column of B .

Second element:

$$(1, 2) = af + bh$$

First row of A multiplied by second column of B .

Third element:

$$(2, 1) = ce + dg$$

Second row of A multiplied by first column of B .

Fourth element:

$$(2, 2) = cf + dh$$

Second row of A multiplied by second column of B .

5. Numerical Example

Multiply:

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

and:

$$B = \begin{bmatrix} 1 & 6 \\ 7 & 8 \end{bmatrix}$$

Step 1: First element

$$\begin{aligned} (2 \times 1) + (3 \times 7) \\ = 2 + 21 = 23 \end{aligned}$$

Step 2: Second element

$$\begin{aligned} (2 \times 6) + (3 \times 8) \\ = 12 + 24 = 36 \end{aligned}$$

Step 3: Third element

$$\begin{aligned} (4 \times 1) + (5 \times 7) \\ = 4 + 35 = 39 \end{aligned}$$

Step 4: Fourth element

$$\begin{aligned} (4 \times 6) + (5 \times 8) \\ = 24 + 40 = 64 \end{aligned}$$

Therefore:

$$AB = \begin{bmatrix} 23 & 36 \\ 39 & 64 \end{bmatrix}$$

6. Important Properties of Matrix Multiplication

1. Non-Commutative Property

Matrix multiplication generally does **not** follow the commutative law.

$$AB \neq BA$$

The order of multiplication matters.

2. Associative Property

Matrix multiplication is associative:

$$(AB)C = A(BC)$$

Changing the grouping does not change the result.

3. Distributive Property

Matrix multiplication distributes over addition:

$$A(B + C) = AB + AC$$

and:

$$(A + B)C = AC + BC$$

4. Identity Matrix Property

The identity matrix behaves like the number 1.

For a 2×2 identity matrix:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Then:

$$AI = A$$

and:

$$IA = A$$

5. Zero Matrix Property

Any matrix multiplied by a zero matrix gives a zero matrix:

$$A \times 0 = 0$$

7. Matrix Multiplication Using Dot Product

Each element of the product matrix is calculated using a dot product.

Example:

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

Multiply corresponding entries:

$$\begin{aligned} &= (1 \times 4) + (2 \times 5) + (3 \times 6) \\ &= 4 + 10 + 18 = 32 \end{aligned}$$

8. Types of Matrix Multiplication

1. Square Matrix Multiplication

Both matrices have equal numbers of rows and columns.

Example:

$$2 \times 2 \text{ multiplied by } 2 \times 2$$

2. Rectangular Matrix Multiplication

Matrices have different dimensions.

Example:

$$(2 \times 3)(3 \times 4)$$

Result:

$$2 \times 4$$

3. Scalar Multiplication

A matrix is multiplied by a single number.

Example:

$$3 \begin{bmatrix} 2 & 4 \\ 1 & 5 \end{bmatrix}$$

Multiply every element by 3:

$$= \begin{bmatrix} 6 & 12 \\ 3 & 15 \end{bmatrix}$$

9. Applications of Matrix Multiplication

Matrix multiplication is used in:

- **Computer graphics** — rotation, scaling, and transformation of images
- **Artificial intelligence and machine learning** — neural network calculations
- **Physics and engineering** — modelling systems and transformations
- **Economics** — input-output models
- **Cryptography** — encoding and decoding information
- **Robotics** — position and movement calculations
- **Statistics** — data analysis

10. Common Mistakes

Mistake 1: Multiplying corresponding elements

Incorrect:

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \times \begin{bmatrix} 3 & 4 \end{bmatrix}$$

Matrix multiplication is **not** the same as multiplying matching positions.

Mistake 2: Ignoring dimension rules

Always check:

$$(m \times n)(n \times p)$$

The two middle dimensions must be equal.

Mistake 3: Assuming the order can be changed

Remember:

$$AB \neq BA$$

for most matrices.

11. Summary

- Matrix multiplication combines matrices using the **row-by-column rule**.
- The number of columns in the first matrix must equal the number of rows in the second matrix.
- The resulting matrix size is determined by the outer dimensions.
- Matrix multiplication is generally **not commutative**.
- It is associative and distributive.
- Matrix multiplication is a fundamental concept used in mathematics, computing, engineering, and science.